READER NOTES

FUNDAMENTALS OF FINANCE | MODULE 7

Bond Portfolio Management (Part 1)

STUDY NOTES · EDITION 1.0 · WEEK 7

How to use these notes

These notes cover the whole of Week 7: every concept and every worked example, written for reading and recall rather than for listening. They are yours to keep, mark up and revise from.

The week has nine topics in three parts. Part A establishes the time value of money and defines the contract a bond represents. Part B prices that contract and reverses the calculation to find its yield. Part C examines what can go wrong and surveys the family of instruments. Read the parts in order the first time; each assumes the last.

When assessment approaches you will not have time to re-read everything, and you will not need to. The Quick Recap boxes, the module summary and the self-check at the back were written for exactly that week. A workable plan: recaps first, then the summary, then attempt the self-check from memory before looking at the answers.

What each topic contains

Every topic follows the same rhythm, so you always know where to look.

Section	What it gives you
In one line	The whole idea compressed to a single sentence
Why this matters	Where the concept sits in the course, and why it earns your attention
The core idea	The full explanation, developed step by step
Figure	A diagram or comparison that makes the idea visible
Callouts	Key ideas, worked examples, pitfalls and exam pointers
Quick recap	Five bullets to revise from

Callouts used in this document

KEY IDEA

The most important takeaway of a section. If you remember nothing else, remember this.

COMMON PITFALL

A mistake learners make repeatedly, in assignments and in examinations.

EXAM POINTER

How the concept is typically tested, and what a full answer must contain.

REAL-WORLD EXAMPLE

A concrete case showing the concept at work outside the classroom.

Contents

Module overview

This module applies one principle — that an asset is worth the present value of its future cash flows — to the instrument where those cash flows are contractually fixed. It begins with the time value of money and the mechanics of discounting, defines the bond contract and its seven features, then prices it and runs the calculation backwards to find yield. It closes on the two risks every bondholder carries and the family of structures built to reshape them.

One thread runs through all nine topics: **because a bond's cash flows are fixed, all the action moves to the discount rate**. The coupon never changes. Everything that happens to a bond's value happens through r.

Learning objectives

By the end of this module, you will be able to:

1. **Explain** the time value of money and apply compounding and discounting to a stream of cash flows.
2. **Describe** the bond contract and compute the price of a bond, including zero coupon and perpetual cases.
3. **Explain** the inverse price-yield relationship and interpret yield to maturity.
4. **Distinguish** interest rate risk from reinvestment risk and classify fixed income instruments.

Topic map

9 topics, three parts. The Source column gives the lecture video each topic draws on, so you can go back to the original; the Serves column shows which objective it supports.

Topic	What it covers	Part	Source	Serves
1	Why a rupee today beats a rupee tomorrow	A · Time value	L24 V1	LO 1
2	Compounding and discounting	A · Time value	L24 V2	LO 1
3	What a bond is	A · The contract	L25 V1	LO 2
4	Pricing a bond	B · Pricing	L25 V2	LO 2
5	The inverse price-yield relationship	B · Pricing	L25 V2	LO 3
6	Yield to maturity	B · Yield	L26 V1	LO 3
7	Interest rate risk and reinvestment risk	C · Risk	L26 V2	LO 4
8	Callable bonds	C · Risk	L26 V3	LO 4
9	The fixed income family	C · Variety	L27 V1	LO 4

Key terms at a glance

Fuller definitions appear in the glossary at the back. Skim these before your first read; they carry most of the week's vocabulary.

Time value of money
The principle that a rupee today is worth more than a rupee tomorrow.

Cash flow equivalence
Bringing every flow to a common date before adding it.

Compounding
Moving money forward in time, with interest earning interest.

Discounting
Moving money back to today; compounding run in reverse.

Bond
A contractual promise to pay interest regularly and return principal on a fixed date.

Debenture
The name a bond carries when issued by a company.

Plain vanilla bond
The simplest structure — regular coupons plus capital at maturity.

Face value
The par amount returned at maturity; also the maturity value.

Coupon rate
The contractual rate setting the cash flows; fixed at issue and never changing.

Discount rate
The market rate used to discount those flows; it changes daily.

Seniority
A bondholder's priority in the payment queue at liquidation.

Annuity
A stream of identical payments, valued in one step.

Zero coupon bond
A bond paying no coupon, bought at a discount and redeemed at face value.

Perpetual bond
A bond with no maturity, paying coupons forever; priced at $C \div r$.

Premium bond
One priced above face value, because r sits below the coupon rate.

Par bond
One priced at face value, because r equals the coupon rate.

Discount bond
One priced below face value, because r sits above the coupon rate.

Risk premium
The extra return demanded over the safe rate for accepting uncertainty.

Yield to maturity
The single discount rate equating a bond's cash flows to its market price.

Interest rate risk
The risk that rising market rates cut a bond's price.

Reinvestment risk
The risk that falling rates reduce the return earned on reinvested coupons.

Duration
A bond's price sensitivity to rate changes; greater for longer maturities.

Callable bond
One the issuer may repay early, at the issuer's sole discretion.

Convertible bond
One the holder may convert into the issuer's shares.

Floating rate bond
One whose coupon resets periodically against a benchmark.

Embedded option
An option living inside a bond, altering both its value and its risk.

PART A · TIME, MONEY AND THE CONTRACT

Comparing rupees across dates

1 Why a Rupee Today Beats a Rupee Tomorrow

PART A · TIME, MONEY AND THE CONTRACT · TOPIC 1 OF 9 · SERVES LEARNING OBJECTIVE 1

IN ONE LINE

Three forces — inflation, opportunity cost and uncertainty — make money available now worth more than the same amount later. Everything in bond pricing follows from that.

Why this matters

This is the foundation of Weeks 7 and 8 entirely. A bond is nothing but a stream of future rupees, so a method for comparing rupees across dates is not background — it is the whole apparatus.

The instinct, and why it is correct

Offered the same ₹10,000 cheque today or in one year — same amount, not a rupee more — almost everyone takes it today. Three forces explain why that instinct is right.

Inflation. Prices creep upward, so each rupee buys a little less. The number in your hand has not shrunk, but its purchasing power has been worn away. A packet of biscuits that cost one amount a few years ago costs more today; the biscuits did not change, the rupee did.

Opportunity cost. Money taken today can be put to work in a fixed deposit or through a platform like Zerodha. A rupee promised next year sits idle and earns nothing in the meantime, and that foregone gain is the cost of waiting.

Uncertainty. A promise is only as good as the promise keeper. Circumstances change and, over a long enough stretch, there is always some risk of default. The money you already hold is certain, and certainty itself carries value.

Cash flow equivalence

Every investment shares the same shape: one outflow today, then a series of inflows over the years that follow. But those inflows sit on different dates, and money on different dates is not the same money.

The golden rule: cash at different dates cannot simply be added. Every flow must first be brought to a common date, normally time zero. The further a rupee sits in the future, the longer its journey back and the smaller it looks once it arrives today.

THREE FORCES, ONE CONCLUSION

INFLATION	OPPORTUNITY COST	UNCERTAINTY
WHAT HAPPENS Prices creep upward over time	WHAT HAPPENS Money held today can be invested	WHAT HAPPENS A promise may not be kept
EFFECT ON THE RUPEE The number is unchanged; the purchasing power is not	EFFECT ON THE RUPEE A rupee promised later earns nothing meanwhile	EFFECT ON THE RUPEE Certain money is worth more than probable money
EXAMPLE The biscuits did not change — the rupee did	EXAMPLE A fixed deposit foregone is a gain foregone	EXAMPLE Over a long horizon, default risk is never zero

Add the three together and you have the ***time value of money*** *— the principle underlying every valuation in this course, applied here to bonds and government securities.*

Figure 1.1 The three forces. Any one of them alone would be enough; together they are decisive.

KEY IDEA

Cash flows on different dates are different *units*, in the way rupees and dollars are different units. Adding them without converting is not an approximation — it is a category error, and the whole machinery of discounting exists to perform the conversion.

EXAM POINTER

Name all three forces with one line each; answers giving only inflation lose two thirds of the marks. Then state the golden rule explicitly — flows on different dates must be brought to a common date before they can be combined.

QUICK RECAP · TOPIC 1

- Three forces make money today worth more: inflation, opportunity cost and uncertainty.
- Inflation erodes purchasing power; the number stays the same but buys less.
- Opportunity cost is the return foregone by waiting.
- Uncertainty means a promise is worth less than possession.
- Cash at different dates cannot be added until brought to a common date, normally time zero.

2 Compounding and Discounting

PART A · TIME, MONEY AND THE CONTRACT · TOPIC 2 OF 9 · SERVES LEARNING OBJECTIVE 1

IN ONE LINE

One formula, run in two directions. Compounding carries a rupee forward; discounting brings it home.

Why this matters

Every bond price in the next seven topics is a discounting exercise. Get this mechanism right once and the rest is bookkeeping.

Compounding

$$CF_t = CF_0 \times (1 + r)^t$$

Growth is not linear. In each period, interest is earned not only on the original amount but on the interest accumulated earlier — that is what makes it *compound* rather than simple. ₹100 invested at 10% grows to ₹133.10 over three years, not ₹130. The extra ₹3.10 is interest earned on interest.

Discounting needs no new equation

It is the same formula rearranged and run backwards: $\mathbf{PV = CF_t \div (1 + r)^t}$. Compounding multiplies by $(1 + r)^t$; discounting divides by it. Mirror images of one operation in opposite directions.

A bond is a bundle of discounted cash flows

Take ABC Ltd: ₹1,000 face value, an 8% coupon, three years. At the end of year one, a coupon of ₹80. At the end of year two, another ₹80. At the end of year three, **₹1,080** — the final coupon and the face value arriving together.

The price today is the present value of every coupon plus the return of principal at maturity. Once you can discount a single future rupee, you can discount a whole stream of them and add the results. That is all pricing a bond asks of you.

ONE FORMULA, TWO DIRECTIONS

	COMPOUNDING	DISCOUNTING
DIRECTION	Moves money forward into the future	Moves money back to today
QUESTION ANSWERED	What will a rupee today grow into?	What is a future rupee worth now?
OPERATION	Multiply by $(1 + r)^t$	Divide by $(1 + r)^t$
RESULT	Future value — larger than the start	Present value — smaller than the future
USED FOR	Growing savings, projecting balances	Pricing bonds, stocks and projects

*₹100 at 10% for three years becomes **₹133.10**, not ₹130. The extra ₹3.10 is interest earned on interest.*

Figure 2.1 Compounding and discounting are the same operation in opposite directions.

COMMON PITFALL

Computing simple interest and calling it compound. Three years at 10% on ₹100 gives ₹30 of simple interest but ₹33.10 compounded. Over long horizons the gap becomes enormous, and bonds are long-horizon instruments.

EXAM POINTER

Write the formula before substituting, and state explicitly that discounting is the same equation rearranged — examiners look for the recognition, not two memorised formulas. For a bond, show that the final cash flow is coupon *plus* face value.

QUICK RECAP · TOPIC 2

- $CF_t = CF_0 \times (1 + r)^t$ — compounding, which earns interest on interest.
- ₹100 at 10% for three years gives ₹133.10, not ₹130.
- $PV = CF_t \div (1 + r)^t$ — discounting is the same formula reversed.
- A bond is a bundle of cash flows, each discounted by its own distance in time.
- The final flow carries the last coupon plus the face value together.

3 What a Bond Is

PART A · TIME, MONEY AND THE CONTRACT · TOPIC 3 OF 9 · SERVES LEARNING OBJECTIVE 2

IN ONE LINE

A bond turns a crowd of small lenders into one very large loan, backed by a binding promise to pay interest and return the principal.

Why this matters

Before pricing a bond you need to know precisely what has been promised. The seven features here determine every calculation that follows.

Three financing doors

A company needs ₹500 crore to build a factory. It has the idea, the land and the engineers, but not the money. It could use **own cash** — but a growing firm rarely has ₹500 crore idle, and emptying the reserves leaves it exposed. It could take a **bank loan** — possible, but leaning on one lender for such a sum is heavy and often expensive. Or it could borrow from **the crowd**: thousands of retail investors, mutual funds, insurers and pension funds each lending a modest amount, which pooled together becomes one very large loan.

The crowd lends to a stranger because the company makes a binding promise: pay interest regularly and return the full principal on a fixed date. That promise is a bond.

Three names, one idea

A **bond** is a binding contract between an issuer who borrows and investors who lend. A **debenture** is the name a bond carries when issued by a company — a corporate bond and a debenture are the same thing. A **fixed income security** is the name earned by its steady income: regular coupons plus capital returned at maturity. The simplest form is the *plain vanilla* bond.

Lending, not owning. A bondholder receives cash returns — interest and principal — but no ownership stake, no vote, and no share of growth. In exchange, they are paid before every shareholder.

How the crowd builds a country

The same act funds far more than factories. Highways, railway lines and airports are frequently financed through bonds rather than a single lender, and national bodies such as the Indian Railway Finance Corporation and the Rural Electrification Corporation raise money this way. In times of crisis, governments have issued war bonds inviting citizens to fund the country directly. Through the RBI's Retail Direct platform, an individual can buy government securities and lend directly to the Government of India.

SEVEN FEATURES DEFINE ANY BOND

Feature	What it specifies	Role
Face value	The par amount returned at maturity	Prices the bond
Coupon rate	The interest rate promised on the face value	Prices the bond
Coupon frequency	How often interest arrives — quarterly, half-yearly, annually	Prices the bond
Maturity date	The day the principal is finally repaid	Prices the bond
Security	Whether the bond is backed by collateral	Describes its safety
Seniority	Priority in the payment queue at liquidation	Describes its standing
Terms and conditions	Call or put options, convertibility, tax treatment, covenants	Describes its structure

*The **first four price the bond**. The last three describe its safety and its standing in the queue.*

Figure 3.1 The seven features. Only four of them enter the pricing formula.

KEY IDEA

A bond is a *contract*, and every term of the contract is negotiable at issue but fixed thereafter. That fixity is exactly what makes the cash flows predictable enough to discount — and what makes the discount rate the only thing that can move.

EXAM POINTER

Name all seven features and, more importantly, say which four price the bond. Distinguish bond, debenture and fixed income security in one line each — a common short-answer question that trips candidates who assume they are three different instruments.

QUICK RECAP · TOPIC 3

- A bond turns many small lenders into one large loan through a binding promise.
- Bond, debenture and fixed income security are three names for the same contract.
- A bondholder lends and is paid before shareholders, but owns nothing and votes on nothing.
- Seven features define any bond; the first four price it.
- Bonds fund infrastructure and government, not just corporate expansion.

PART B · PRICING AND YIELD

What the contract is worth, and what it yields

4 Pricing a Bond

PART B · PRICING AND YIELD · TOPIC 4 OF 9 · SERVES LEARNING OBJECTIVE 2

IN ONE LINE

The price is the present value of every coupon plus the principal — and the annuity shortcut turns many calculations into two.

Why this matters

This is the central calculation of Week 7. Everything in Part C is an examination of what happens to this number when the discount rate moves.

Three ways to write the same thing

The general formula: $P_0 = \Sigma\,[\,C_t \div (1 + r)^t\,]$, summed over every period from t = 1 to T.

The debenture formula: with equal coupons every year, $P_0 = C \div (1+r)^1 + C \div (1+r)^2 + \ldots +$ **(F + C)**$\div(1+r)^T$. The detail to notice is the final term: F plus C, not just C, because principal returns with the last coupon.

The single annuity approach: identical coupons form an annuity, so the price can be found in two pieces instead of many. $P_0 = C \times [1 - (1 + r)^{-T}] \div r + M \div (1 + r)^T$ — the present value of all coupons as one annuity, plus the present value of the maturity value as a lump sum.

Two special cases

A **zero coupon bond** has C = 0, so the annuity part vanishes entirely and price = $M \div (1 + r)^T$. You buy at a discount today and receive full face value at maturity, with nothing in between.

A **perpetual bond** has T tending to infinity, and the price collapses to $P_0 = C \div r$. Coupons forever, valued in a single fraction.

Coupon rate versus discount rate

These are constantly confused and must never be blurred. The **coupon rate** sets the cash flows you receive; it is the nominal or contractual rate, fixed at issue, and it never changes. The **discount rate** is what you discount those flows by; it is the market rate, yield or opportunity cost, and it moves every day.

One decides what you are owed. The other decides what that is worth now. Note also that the Capital Asset Pricing Model is *not* used to find the discount rate for bonds.

ONE BOND, PRICED TWO WAYS · THE ANNUITY ROUTE IS FASTER

Step	Working	Result
Inputs	Face ₹1,000 · coupon 10% so C = ₹100 · required return r = 8% · T = 5 years, annual	—
Coupons as an annuity	$100 \times [1 - (1.08)^{-5}] \div 0.08 = 100 \times 3.9927$	₹399.27
Principal as a lump sum	$1{,}000 \div (1.08)^{5}$	₹680.58
Price	399.27 + 680.58	**₹1,079.85**

r = 8% sits ***below*** *the 10% coupon, so the price lands* ***above*** *face value — a premium bond, exactly as the rule in Topic 5 predicts.*

Figure 4.1 The annuity approach. Two present values instead of five.

COMMON PITFALL

Forgetting that the final cash flow is the coupon *plus* the face value. Discounting only the coupon in the last period understates the price by the present value of the entire principal — usually the largest single component.

EXAM POINTER

Use the annuity approach and show both components separately before adding — the coupon annuity and the discounted face value. Then state whether the result is a premium, par or discount bond and why. Both special cases are examinable: zero coupon drops the annuity term, perpetual collapses to C ÷ r.

QUICK RECAP · TOPIC 4

- $P_0 = \Sigma[C_t \div (1 + r)^t]$ — the present value of every cash flow.
- The final term is (F + C), because principal returns with the last coupon.
- The annuity approach computes coupons in one step plus the discounted face value.
- A ₹1,000 face, 10% coupon, five-year bond at r = 8% is worth ₹1,079.85.
- Zero coupon: $P_0 = M \div (1 + r)^T$. Perpetual: $P_0 = C \div r$.

5 The Inverse Price-Yield Relationship

PART B · PRICING AND YIELD · TOPIC 5 OF 9 · SERVES LEARNING OBJECTIVE 3

IN ONE LINE

Price and yield sit on a seesaw and can never rise together. This is the single most important picture in fixed income.

Why this matters

Everything about bond risk in Part C and the whole of Week 8 depends on this one relationship. It is also the idea most often stated correctly and understood shallowly.

The four drivers of price

Coupon (C) — the annual interest paid. Higher coupon, higher price. **Face value (M)** — the amount returned at maturity. Higher face, higher price. **Maturity (T)** — the years remaining. Longer, all else equal, higher price. **Discount rate (r)** — the required return or market yield. Higher r, **lower** price.

Only the last one runs the other way. Hold C, M and T fixed, change only r, and that isolation reveals the price-yield relationship.

The seesaw

Picture a children's seesaw. On one end sits the price of a bond; on the other sits the level of market interest rates. The defining feature of a seesaw is that the two ends can never rise together.

When rates rise, prices fall. Priya holds a bond paying 8%. The RBI raises rates to 10%, so fresh bonds now pay 10% and Priya's older bond looks stingy. To sell it, she must drop its price. When rates fall to 6%, her 8% bond looks generous beside the new crowd, buyers compete, and its price rises.

Nothing about the bond changed — only the rates around it. Competition between old and new bonds is the entire engine.

The three zones

Where r sits relative to the coupon rate determines everything. When **r is below the coupon**, the price sits above face value — a *premium* bond. When **r equals the coupon**, price equals face — a *par* bond. When **r is above the coupon**, price sits below face — a *discount* bond. This is a mathematical certainty, not a tendency.

ONE BOND, FIVE DISCOUNT RATES · ₹1,000 FACE, 10% COUPON, FIVE YEARS

Discount rate r	Price	Zone	Because
6%	₹1,168.49	Premium	r below the 10% coupon
8%	₹1,079.85	Premium	r below the coupon
10%	₹1,000.00	Par	r equals the coupon exactly
12%	₹927.90	Discount	r above the coupon
14%	₹862.68	Discount	r above the coupon

*A discount bond is **not cheaper** or a better deal than a premium bond. Each price is fair; it simply reflects a different discount rate applied to the same machinery.*

Figure 5.1 The price-yield relationship for one bond. Only r changes across the rows.

KEY IDEA

The seesaw is not a market convention or a behavioural tendency. It is arithmetic: r sits in the denominator of every term in the pricing formula, so raising it must lower every present value and therefore the price.

COMMON PITFALL

Treating a bond trading at a discount as a bargain. The price is not good or bad — it is what those cash flows are worth at the required return of the day. A discount bond and a premium bond can offer identical value.

EXAM POINTER

Name all four drivers and identify r as the only one moving inversely. State the three zones as a rule linking r to the coupon rate, and be ready to work backwards — given that a bond trades at a premium, you can conclude immediately that r is below the coupon.

QUICK RECAP · TOPIC 5

- Four drivers: coupon, face value, maturity and the discount rate.
- Only the discount rate moves price in the opposite direction.
- Price and market rates sit on a seesaw and can never rise together.
- r below coupon gives a premium; r equal gives par; r above gives a discount.
- A discount bond is not cheaper — both prices are fair at their respective rates.

6 Yield to Maturity

PART B · PRICING AND YIELD · TOPIC 6 OF 9 · SERVES LEARNING OBJECTIVE 3

IN ONE LINE

The single discount rate that makes a bond's cash flows equal its market price — the equation run backwards.

Why this matters

Topics 4 and 5 took r as the input and produced a price. In real markets the price is observed and r is the unknown. This topic reverses the calculation, and Week 8 develops it further.

The risk premium first

Why would anyone pay more to one borrower than another? Imagine a savings account offering 6% and a risky debenture also offering 6%. Same return, more risk — nobody would lend to the company. The issuer must raise its rate to attract anyone: 7%, 8%, 9%.

That extra slice paid over the safe rate is the **risk premium**. Expected return = risk-free rate + risk premium. Safer bond, thinner premium; riskier borrower, fatter premium. There is no free lunch.

The definition

Yield to maturity is the bond's internal rate of return — the single discount rate that makes the present value of all its future cash flows exactly equal to its current market price.

$$P_0 = C \times [1 - (1 + YTM)^{-N}] \div YTM + M \div (1 + YTM)^N$$

YTM sits in the denominator underneath every cash flow, shrinking each one back to today. Because it lives there, the price-yield seesaw is built into the formula itself: raise YTM and every present value drops.

What YTM folds together: coupon income collected along the way, any capital gain or loss booked when the bond matures at face value, and the time value of money governing every flow in between — three things in one number.

Price tells you the yield

Trading at **par**, YTM equals the coupon rate. Trading at a **discount**, YTM exceeds the coupon rate — you pay less than face, so you earn more than the coupon rate. Trading at a **premium**, YTM is below the coupon rate, because paying more than face squeezes the effective yield down.

A lower purchase price always means a higher yield to maturity. Buy a 10% coupon bond at ₹95 and its YTM exceeds what it would be at ₹102 — both still return ₹100 of face value at maturity. Pay less today, earn more in total.

YIELD CANNOT BE SOLVED BY HAND · THE TWO SPREADSHEET ROUTES

Function	When to use it	Worked example	Result
=YIELD(settlement, maturity, rate, price, redemption, frequency)	When settlement and maturity *dates* are known	Settlement 1 Jan 2024, maturity 1 Jan 2027, coupon 10%, price 95, redemption 100, frequency 1	**12.08%**
=RATE(nper, pmt, pv, fv)	When only the number of *periods* is known	=RATE(3, 8, −95, 100) — note the present value is entered as a negative, being a cash outflow	The same rate
=PRICE(settlement, maturity, rate, yield, redemption, frequency)	The reverse direction — yield in, price out	The same bond at a yield of 12.08%	**95.01**

YIELD and PRICE run the ***same relationship in opposite directions****. Feed the output of one into the other and you return to where you started.*

Figure 6.1 Computing yield to maturity. It requires iteration, which is why a spreadsheet is not optional.

COMMON PITFALL

Reporting YTM as though it were guaranteed. It rests on two assumptions: that you hold to maturity, and that every coupon is reinvested at the YTM rate. The second is the demanding one — if rates fall after a coupon arrives, you cannot reinvest at the old rate, and your realised return drifts below the promised figure.

EXAM POINTER

Define YTM as the internal rate of return equating cash flows to price, then name the three things it folds together. State the par, discount and premium relationships as a rule. If asked why YTM needs a spreadsheet, the answer is that it cannot be solved algebraically — it requires iteration.

QUICK RECAP · TOPIC 6

- Expected return = risk-free rate + risk premium; riskier borrowers pay more.
- YTM is the single discount rate equating a bond's cash flows to its market price.
- It combines coupon income, capital gain or loss, and the time value of money.
- At par YTM equals the coupon; at a discount it exceeds it; at a premium it is below.
- YTM requires iteration — use =YIELD or =RATE; =PRICE runs the same relationship backwards.

PART C · RISK AND VARIETY

What can go wrong, and how structures respond

7 Interest Rate Risk and Reinvestment Risk

PART C · RISK AND VARIETY · TOPIC 7 OF 9 · SERVES LEARNING OBJECTIVE 4

IN ONE LINE

Two risks, opposite triggers, one root cause. Whichever way rates move, one of them is at work.

Why this matters

This pair is the entire basis of the immunisation strategy in Week 8. Understanding that they move in opposite directions is what makes balancing them possible.

Interest rate risk

Definition: the risk that a change in market interest rates makes a bond's price fluctuate. The price moves not because the issuer fails, but because rates move.

Maturity multiplies it. The inverse rule tells you the *direction* of the move, not its *size*. Distant cash flows are hit hardest when the discount rate changes, so a thirty-year bond reacts far more violently than a two-year one. This sensitivity has a name — *duration* — and Week 8 measures it.

Silicon Valley Bank in March 2023 is the cautionary case. The bank held a large pile of long-term, low-coupon bonds. Through 2022 rates rose sharply, pushing those bonds' value steeply down. Held quietly, the losses stayed on paper. When the bank was forced to sell to raise cash, the paper losses turned painfully real — and the seesaw toppled an entire bank.

Reinvestment risk

The quieter risk lives not in the price but in the coupons. Yield to maturity quietly assumes every coupon is reinvested at that same YTM. But if rates fall from 10% to 5%, those coupons earn only 5%, and the realised return drops below the promised yield. That shortfall is **reinvestment risk**.

The mirror image

The two are triggered by opposite movements. Interest rate risk bites when rates *rise*; reinvestment risk bites when rates *fall*. They share one root cause — market rates change over time — and one of them is always operating.

That opposition is not a nuisance. It is an opportunity, and Week 8 exploits it: if the two effects can be made to offset, a portfolio can be protected against rate movements in either direction.

TWO RISKS, OPPOSITE TRIGGERS

	INTEREST RATE RISK	REINVESTMENT RISK
WHAT IT HITS	The bond's market value — the price	The return earned on coupons
TRIGGERED WHEN	Market rates rise	Market rates fall
MECHANISM	The old bond becomes less attractive, so its price drops	Coupons are reinvested at a lower rate
FELT BY	Anyone selling before maturity	Anyone reinvesting coupons
LINKED IDEA	Duration — maturity sensitivity	The yield to maturity assumption

One bites when rates climb, the other when they fall. ***Opposite triggers, same root cause*** *— and that opposition is what makes immunisation possible in Week 8.*

Figure 7.1 The two bond risks. Note that they can never both bite at once.

KEY IDEA

Because the two risks are triggered by opposite movements, a bondholder is never fully exposed to both simultaneously. That is not a coincidence — it is the structural feature that lets a portfolio be immunised by balancing one against the other.

EXAM POINTER

Answer on five axes: what each risk hits, what triggers it, the mechanism, who feels it, and the linked concept. State explicitly that they have opposite triggers and a common cause — that sentence is usually worth a mark on its own, and it sets up the Week 8 material.

QUICK RECAP · TOPIC 7

- Interest rate risk is the price fluctuation caused by changing market rates.
- It grows with maturity, because distant cash flows are hit hardest — the idea behind duration.
- SVB in March 2023 shows what happens when paper losses are forced into realisation.
- Reinvestment risk is the shortfall when coupons are reinvested at lower rates than YTM assumed.
- The two have opposite triggers and one root cause; whichever way rates move, one is at work.

8 Callable Bonds

PART C · RISK AND VARIETY · TOPIC 8 OF 9 · SERVES LEARNING OBJECTIVE 4

IN ONE LINE

The issuer may repay early, at its sole discretion — and it will choose precisely the moment the investor least wants it to.

Why this matters

An embedded option changes both the value and the risk of a bond, and the callable bond is the cleanest illustration of that. It also previews the option logic of Weeks 9 and 10.

The refinancing instinct

Rahul takes a home loan at 12%. A few years later rates fall and new loans are available at 8%. The first thing he wants to do is repay the expensive loan and borrow again more cheaply — refinancing. Companies think exactly the same way, and that idea is the callable bond.

Definition: a callable bond gives the issuer the right to repay it before its stated maturity date, handing back the principal early, closing the contract and walking away from the remaining coupons.

The asymmetry

The issuer decides if and when to call, and the decision is final. The investor can neither force early repayment nor prevent it. The clause quietly hands the steering wheel to the borrower, and every other feature of the bond flows from that.

Why the call always arrives at the worst moment

The engine underneath is the inverse relationship from Topic 5. When rates fall, an old high-coupon bond becomes prized — investors compete to buy it and its price climbs. That is precisely the moment the investor most wants to keep collecting those payments, and precisely the moment the issuer wants to snatch it back.

So the issuer calls, the principal is returned, the high coupons stop, and the investor is pushed back into a lower-paying market. This is not a neutral feature: the call option works **systematically in favour of the issuer** and against the investor, who loses exactly the payments that had become most valuable.

Why anyone buys one

Think of a tenant who wants the freedom to end a lease early whenever it suits them. The landlord, bearing that uncertainty, asks a higher rent in return. The same logic carries across: the issuer holds the flexibility, so the issuer must pay for it — through a higher coupon, a lower issue price, or both.

The rule: a callable bond offers a higher yield than an identical non-callable bond. The extra yield is the rent the issuer pays for the right to walk away early.

AN EMBEDDED OPTION, AND WHO IT FAVOURS

RATES FALL
New borrowing becomes cheaper for the issuer

›

THE BOND'S PRICE CLIMBS
Its high coupon is now prized · the investor wants to hold

›

THE ISSUER CALLS
Principal returned early · the high coupons stop

›

THE INVESTOR REINVESTS
Back into a lower-paying market · exactly what they did not want

*A callable bond is not just a bond — it is a bond with an **option living inside it**. That option changes its value, since you pay less for something that can be taken away, and changes its risk.*

Figure 8.1 The call sequence. Every step is favourable to the issuer and adverse to the investor.

KEY IDEA

The higher yield on a callable bond is not generosity and not a free lunch. It is *compensation* for having sold the issuer a valuable option — which is why comparing a callable bond's yield with a non-callable one, without adjusting, is meaningless.

EXAM POINTER

Define the call feature and stress that the decision is the issuer's alone. Explain the timing asymmetry — the call arrives when rates fall, which is when the investor most wants to hold — and close with the compensation rule: callable bonds must yield more than otherwise identical non-callable bonds.

QUICK RECAP · TOPIC 8

- A callable bond lets the issuer repay before maturity, at the issuer's sole discretion.
- The investor can neither force nor prevent the call.
- Issuers call when rates fall — exactly when the investor most wants to keep the high coupon.
- The option works systematically in the issuer's favour.
- In compensation, a callable bond must offer a higher yield than an identical non-callable bond.

9 The Fixed Income Family

PART C · RISK AND VARIETY · TOPIC 9 OF 9 · SERVES LEARNING OBJECTIVE 4

IN ONE LINE

Six structures, one underlying idea — every member is a loan, but each is shaped for a different need.

Why this matters

The closing topic of the week. Sorting instruments by their features is what lets you predict how each will behave when rates move, which is where Week 8 begins.

One family, many structures

Walk into a large showroom and at first glance every vehicle appears to be a car. On closer inspection some are hatchbacks, some sedans, some SUVs, some electric. Fixed income is the same.

Government bonds fund public spending on roads, railways, schools and hospitals, and are among the safest investments because the sovereign stands behind them — T-bills, dated G-secs and State Development Loans. **Corporate bonds** fund expansion and working capital; a company can struggle in ways a sovereign rarely does, so these carry genuine credit risk and must offer a higher yield. **Zero coupon bonds** pay nothing annually — the return lies entirely in the price gap, paying ₹70 today for ₹100 in five years. **Callable bonds** let the issuer repay early. **Convertible bonds** begin as ordinary bonds but let the holder convert into shares, giving downside protection with equity upside — the only security that turns debt into ownership. **Floating rate bonds** reset their coupon against a benchmark, so when the repo rate moves the coupon moves with it and the price barely swings.

Ranked by rate sensitivity

Duration drives the reaction to rates — the further away the cash flows, the harder the price swings. **Floating rate** bonds are steadiest, because the coupon resets. **Convertibles** follow, since the equity option softens rate moves. **Government** and **corporate** bonds sit in the middle with fixed coupons and long duration, the latter carrying credit risk as well. **Callable** bonds add the issuer's option. **Zero coupon** bonds react most violently of all, because their entire value sits in one distant payment.

Five questions to ask of any bond

Who issues it — a government or a company? How risky is it — how likely is this borrower to repay in full and on time? What cash flows arrive, and exactly when? Is it callable — can the issuer repay early and take away my high coupon? And what yield am I earning for the risk I am actually taking?

SIX SECURITIES, SEVEN FEATURES

Type	Issuer	Coupon	Credit risk	Early redemption	Equity conversion	Rate sensitivity
Government	Govt	Fixed	Very low	No	No	High
Corporate	Company	Fixed	Moderate to high	No	No	High
Zero coupon	Either	None	Depends	No	No	Very high
Callable	Company	Fixed	Moderate	Yes	No	High
Convertible	Company	Fixed	Moderate	No	Yes	Moderate
Floating rate	Either	Variable	Depends	Usually no	No	Lower

*Read the last column as a **ranking**. Zero coupon bonds swing hardest because all their value sits in one distant payment; floating rate bonds barely move because the coupon resets.*

Figure 9.1 The fixed income family. The rate sensitivity column previews the whole of Week 8.

KEY IDEA

Every variation in this family is a variation in *when* the cash arrives or *who* controls the timing. Fix that and you can predict the instrument's behaviour without memorising it.

EXAM POINTER

Name all six types with the defining feature of each. If asked to rank by rate sensitivity, give the order and the reason — duration, driven by how distant the cash flows are. The zero coupon and floating rate extremes are the two that must be right.

QUICK RECAP · TOPIC 9

- Six structures: government, corporate, zero coupon, callable, convertible, floating rate.
- Government bonds are safest; corporate bonds pay a default premium for credit risk.
- Zero coupon bonds earn entirely through the price gap and are the most rate-sensitive.
- Convertibles are the only security that turns debt into ownership.
- Floating rate bonds reset their coupon, so their price barely moves when rates do.

Module summary, by learning objective

Organised the way assessment will test you, rather than in topic sequence.

LO	What you should now be able to do	Where it was covered
1	Explain the time value of money and apply compounding and discounting	Topics 1 and 2 · Figures 1.1, 2.1
2	Describe the bond contract and compute a bond price	Topics 3 and 4 · Figures 3.1, 4.1
3	Explain the price-yield relationship and interpret yield to maturity	Topics 5 and 6 · Figures 5.1, 6.1
4	Distinguish the two bond risks and classify fixed income instruments	Topics 7, 8 and 9 · Figures 7.1, 8.1, 9.1

The module in six sentences

Three forces — inflation, opportunity cost and uncertainty — make a rupee today worth more than a rupee tomorrow, so cash flows on different dates cannot be added until brought to a common date. Compounding carries money forward and discounting brings it back, using the same formula in opposite directions. A bond turns a crowd of small lenders into one large loan through a binding contract, defined by seven features of which four price it. Its price is the present value of every coupon plus the principal, most easily computed by treating the coupons as an annuity and the face value as a lump sum. Price and yield sit on a seesaw and can never rise together, so where the discount rate sits relative to the coupon determines whether the bond trades at a premium, at par or at a discount — and yield to maturity is that same equation run backwards from an observed price. Two risks follow, triggered by opposite rate movements, and the family of structures from zero coupon to floating rate exists precisely to reshape how strongly a bond responds to them.

HOW THE NINE TOPICS CONNECT

PART A · TIME AND CONTRACT

1 · Three forces
2 · Compounding
3 · What a bond is

TAKEAWAY
Rupees on different dates differ

PART B · PRICING AND YIELD

4 · Pricing
5 · The seesaw
6 · Yield to maturity

TAKEAWAY
All the action is in r

PART C · RISK AND VARIETY

7 · Two risks
8 · Callable bonds
9 · The family

TAKEAWAY
Structures reshape the response

Because a bond's cash flows are fixed, all the action moves to the discount rate. *The coupon never changes; everything that happens to a bond's value happens through r.*

Figure M1 How the nine topics connect. Revise along the parts, not down the list.

Glossary and self-check

Term	Definition
Time value of money	The principle that a rupee today is worth more than a rupee tomorrow.
Cash flow equivalence	Bringing every flow to a common date before adding it.
Compounding	Moving money forward in time, with interest earning interest.
Discounting	Moving money back to today; compounding run in reverse.
Bond	A contractual promise to pay interest regularly and return principal on a fixed date.
Debenture	The name a bond carries when issued by a company.
Plain vanilla bond	The simplest structure — regular coupons plus capital at maturity.
Face value	The par amount returned at maturity; also the maturity value.
Coupon rate	The contractual rate setting the cash flows; fixed at issue and never changing.
Discount rate	The market rate used to discount those flows; it changes daily.
Seniority	A bondholder's priority in the payment queue at liquidation.
Annuity	A stream of identical payments, valued in one step.
Zero coupon bond	A bond paying no coupon, bought at a discount and redeemed at face value.
Perpetual bond	A bond with no maturity, paying coupons forever; priced at $C \div r$.
Premium bond	One priced above face value, because r sits below the coupon rate.
Par bond	One priced at face value, because r equals the coupon rate.
Discount bond	One priced below face value, because r sits above the coupon rate.
Risk premium	The extra return demanded over the safe rate for accepting uncertainty.
Yield to maturity	The single discount rate equating a bond's cash flows to its market price.
Interest rate risk	The risk that rising market rates cut a bond's price.
Reinvestment risk	The risk that falling rates reduce the return earned on reinvested coupons.
Duration	A bond's price sensitivity to rate changes; greater for longer maturities.
Callable bond	One the issuer may repay early, at the issuer's sole discretion.
Convertible bond	One the holder may convert into the issuer's shares.

Term	Definition
Floating rate bond	One whose coupon resets periodically against a benchmark.
Embedded option	An option living inside a bond, altering both its value and its risk.

Self-check

Ungraded retrieval practice. Answer from memory first, then check.

Q1 ₹100 is invested at 10% for three years. Why is the result ₹133.10 rather than ₹130?

> **Because interest compounds.** $CF_t = 100 \times (1.10)^3$ = ₹133.10. Simple interest would give ₹30; the extra ₹3.10 is interest earned on interest already credited.

Q2 A ₹1,000 face value bond has a 10% coupon and five years to maturity. The required return is 8%. Price it, and say which zone it falls in.

> **₹1,079.85 — a premium bond.** Coupons as an annuity: 100 × 3.9927 = ₹399.27. Principal: $1{,}000 \div 1.08^5$ = ₹680.58. Because r of 8% sits below the 10% coupon, the price must exceed face value.

Q3 A bond trades at a discount. What does that tell you about its yield to maturity relative to its coupon rate?

> **YTM exceeds the coupon rate.** You pay less than face value but still receive the full face value at maturity, so the capital gain lifts your total return above the coupon rate. A lower purchase price always means a higher YTM.

Q4 Rates rise sharply. Which of the two bond risks is at work, who feels it, and which risk is dormant?

> **Interest rate risk, felt by anyone selling before maturity.** Prices fall because newer bonds offer more. Reinvestment risk is dormant — in fact coupons now reinvest at higher rates, which is favourable. The two have opposite triggers.

Q5 Why must a callable bond offer a higher yield than an otherwise identical non-callable bond?

> **Because the investor has sold the issuer a valuable option.** The issuer will call when rates fall — exactly when the high coupon is most valuable to hold — so the investor bears an adverse, one-sided risk and must be compensated through a higher coupon, a lower issue price, or both.

Q6 Rank a zero coupon bond and a floating rate bond by sensitivity to interest rate changes, and explain why.

> **Zero coupon is most sensitive; floating rate is least.** A zero coupon bond's entire value sits in one distant payment, so a change in the discount rate is applied over the full horizon. A floating rate bond resets its coupon with the benchmark, so its price barely moves at all.